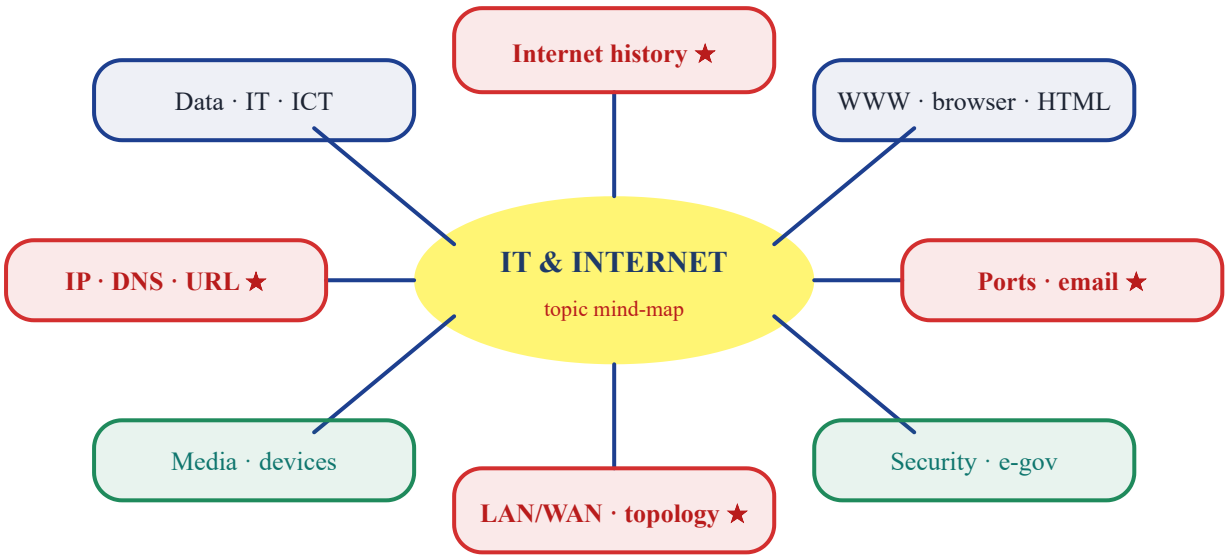
IT & the Internet

IT/ICT · Internet history · IP · DNS · URL · Email · Networks · Cyber security · E-governance: exam-ready in 11 sheets



↑ the whole topic on one page: red = high-yield, green = applied

Syllabus coverage

"Introduction to Computer Systems: … Hardware … Software … Basic architecture of a computer; **Basic Information Technology**; **The Internet**; Number system …". This notebook covers IT/ICT, the Internet and its services, addressing, network basics, cyber security and e-governance at MCQ depth. Layers, protocol internals, subnetting and cryptography are in Chapter 9 (Networking & Security).

High-yield list

① ARPANET 1969 · TCP/IP 1983
② WWW: Berners-Lee, 1989
③ IPv4 32 bit · IPv6 128 bit
④ DNS: name → IP
⑤ port numbers 21…443
⑥ mesh: n(n−1)/2 links

Legend: **yellow** = key term · **red circle** = must-memorise fact · **purple box** = memory trick · **red box** = exam trap · **green** = worked example

Contents

✓ Data, IT, ICT & e-governance …… p. 2
✓ Internet: history & Bangladesh …… p. 3
✓ How it works: IP, domain, DNS … p. 4
✓ WWW, URL, services & ports … p. 5
✓ Email …… p. 6
✓ Network types & topologies …… p. 7
✓ Media, devices, connectivity …… p. 8
✓ Cyber security & e-commerce … p. 9
✓ Night-before sheet …… p. 10
✓ Practice MCQs + written …… p. 11

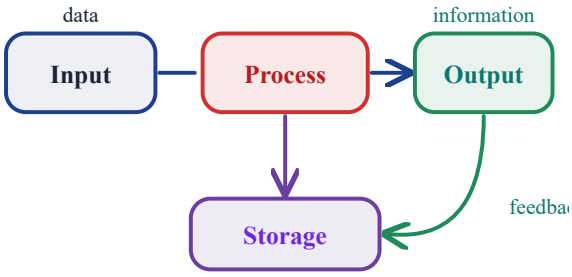
1 Data, Information & IT

Data: raw, unorganised facts (marks, names) with no meaning on their own.

Information: data **processed** into a meaningful, useful form.

IT: **Information Technology**: using computers, storage, networks and software to **create, process, store, secure and exchange** electronic data.

ICT: **Information & Communication Technology** = IT + communication technology (telephone, mobile, broadcasting, Internet). The **wider term**, used in Bangladesh policy (ICT Division, ICT Act 2006).



the data-processing cycle (IPO + storage)

Components of an IT system: hardware · software · data · network · people (+ procedures in some books).

Functions of IT: capture → process → store → retrieve → transmit, plus **automation**, decision support and security.

Memory trick: **DIKW** pyramid: **D**ata → **I**nformation → **K**nowledge → **W**isdom. Each step up adds meaning.

Good information is accurate, timely, relevant, complete and concise. Uses: banking, e-learning, telemedicine, e-commerce (p. 9), government ↓.

2 E-governance & Digital Bangladesh

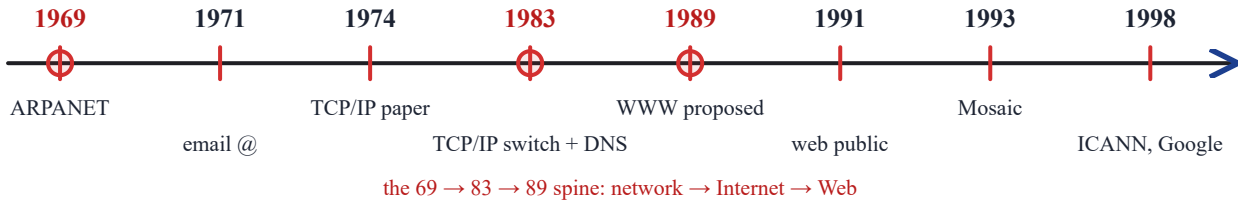
E-governance: using ICT to deliver government services and information: **transparent, faster, cheaper**, fewer office visits. Models: **G2C** (citizens), **G2B** (business), **G2G** (agencies), **G2E** (employees).

Initiative	What to remember
BCC	Bangladesh Computer Council (Act 1990), govt ICT body
ICT Act 2006	legal e-records and **digital signatures**
Digital Bangladesh	announced 2008, target **2021** → Smart Bangladesh 2041
a2i · UDC	a2i (2007) · UDCs (2010): **one-stop rural service**
Systems	bangladesh.gov.bd, e-Nothi, e-GP (2011), NID
Helplines	**999** emergency (2017) · 333 info & services

Exam trap: Data = **input**, information = **output** (and can become the next system's data). ICT ⊃ IT. E-governance ≠ a govt website.

3 The Internet: a Network of Networks

Definition: The Internet is the global, public **network of networks** whose computers exchange data with the **TCP/IP protocol suite**. Nobody owns it; it is coordinated by **ICANN/IANA** (names & numbers), **IETF** (standards, published as RFCs), **W3C** (web standards) and **ISOC**.



Year	Milestone	Name to remember
1969	**ARPANET** (US DoD ARPA): UCLA → SRI, 29 Oct	4 nodes: UCLA, SRI, UCSB, Utah
1971	first network email, the @ sign	Ray Tomlinson
1974	TCP/IP design published	**Vint Cerf & Bob Kahn**
1983	1 Jan: NCP → TCP/IP = **Internet born**; DNS	DNS: Paul Mockapetris
1989	**WWW proposed** at CERN (browser 1990)	**Tim Berners-Lee**
1993	Mosaic: first popular graphical browser	NCSA, Marc Andreessen
1994/98	W3C (1994) · ICANN, Google (1998)	Google: Page & Brin

Bangladesh milestones:

1993: offline email (UUCP dial-up)
4 June 1996: **online Internet via VSAT**
21 May 2006: **SEA-ME-WE 4** submarine cable, Cox's Bazar
2017: SEA-ME-WE 5 at Kuakata (BSCCL)
Mobile: 3G 2012 · 4G 2018 · 5G trial 2021 (Teletalk)
12 May 2018: Bangabandhu Satellite-1 launched
Regulator: **BTRC** (2002)

Web 1.0 read-only static pages → **Web 2.0** read-write (blogs, wikis, social media) → **Web 3.0** semantic / decentralised.

Memory trick: **69-83-89**: ARPANET is born (69), it becomes the Internet (83), the Web is proposed on it (89).

Exam trap: **Internet ≠ WWW.** The Internet is the infrastructure (1969/1983); the Web is one service running on it (1989). Email (1971) is older than the Web. "Father of the Internet" = Cerf (& Kahn); "father of the Web" = Berners-Lee.

4 How Data Crosses the Internet

Packet switching: a message is cut into **packets**, each carrying source and destination **IP addresses**. **Routers** forward them hop by hop on the best route they know, so packets may take **different routes** and arrive out of order; **TCP** reorders them and resends lost ones, while **IP** only delivers (best effort). **Path in Bangladesh:** submarine cable / ITC → **IIG** → NIX / NTTN → **ISP** → user, all licensed by BTRC.

5 IP Address, Domain Name & DNS

	IPv4	IPv6	MAC address
Size	**32 bits** (4 bytes)	**128 bits** (16 bytes)	48 bits (6 bytes)
Written as	4 decimal octets, dots	8 groups of 4 hex digits, colons	6 hex pairs
Example	192.168.1.10	2001:db8::1	00:1A:2B:3C:4D:5E
Total / layer	$2^{32} \approx 4.3$ billion · L3	$2^{128} \approx 3.4 \times 10^{38}$ · L3	burnt into the NIC · L2

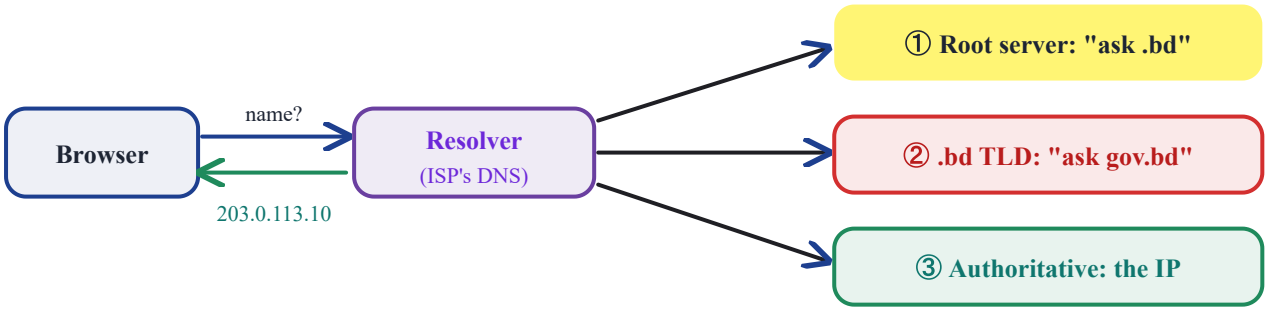
Class	1st octet	Use
A	1–126	huge networks
B	128–191	medium
C	192–223	small
D	224–239	multicast
E	240–255	experimental

Private (not routed on the Internet): 10.x.x.x · 172.16–31.x.x · **192.168.x.x**

Loopback: 127.0.0.1 (localhost)

Static IP is fixed by hand; **dynamic** IP is leased by **DHCP**. Subnetting → Chapter 9.

Domain name: a readable name for an IP address, read **right to left**: root (.) → **TLD** → second level → host. **gTLD:** .com .org .net .edu .gov .int · **ccTLD:** 2 letters, **.bd** (run by BTCL), .in, .uk; .বাংলা is the Bangla IDN ccTLD.



DNS resolution of www.example.gov.bd: the resolver asks root → TLD → authoritative (UDP port 53)

Memory trick: DNS is the Internet's **phone book**: you know the name, it finds the number. DHCP is the **hotel desk** that hands out room numbers.

Exam trap: DNS translates names → IP; **DHCP** assigns IPs; **ARP** finds the MAC for an IP. 127.0.0.1 is **not** a private-range address: it is loopback.

6 World Wide Web (WWW)

Definition: The Web is a system of interlinked **hypertext** documents on the Internet. Three inventions by Berners-Lee hold it together: **HTML** (page language), **HTTP** (transfer protocol) and **URL** (address). A **browser** requests pages from a **web server** and renders them.



URL = Uniform Resource Locator · host = www.example.gov.bd

Term	Meaning	Term	Meaning
Web page	one HTML document	Website	pages under one domain
Home page	a site's first/main page	Hyperlink	link to another resource
Web server	serves pages (Apache, IIS)	Browser	Chrome, Firefox, Edge
Search engine	indexes the web (Google)	Cookie	small text file in the browser

HTTP is **stateless** (cookies/sessions add memory). **HTTPS** = HTTP over **TLS** (formerly SSL), padlock, port 443. Static page = same for all; dynamic = built per request. First browser *WorldWideWeb* (1990); first popular graphical one **Mosaic** (1993); first search engine **Archie** (1990).

7 Internet Services & Well-known Ports

Service	Protocol	Port
File transfer	FTP	**20/21**
Secure login	SSH	22
Remote login	Telnet	23
Send email	SMTP	25
Name lookup	DNS	53
Web	HTTP	**80**
Read email	POP3 · IMAP	110 · 143
Secure web	HTTPS	**443**

Memory trick: The **20s** count up: FTP 21 → SSH 22 → Telnet 23 → (skip) → SMTP 25. Then DNS 53, web 80, POP 110, IMAP 143, secure web 443.

Other services: VoIP calls, video conferencing (Zoom), chat, streaming, social networking, e-learning, IoT (term: Kevin Ashton, 1999).

Cloud: SaaS (Gmail, Google Docs) · PaaS (App Engine) · IaaS (AWS EC2).

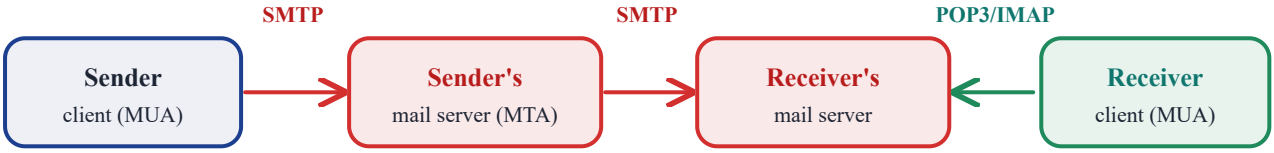
Exam trap: FTP uses **two** ports: 21 = control, 20 = data. HTTPS is not a new protocol: it is HTTP carried inside TLS. A domain name is only **one part** of a URL, and a port is written after a colon (:443).

8 Electronic Mail (Email)

Definition: Email sends digital messages (with attachments) between mailboxes through mail servers. First network email: **Ray Tomlinson, 1971**, who chose @ to separate the user from the host. Hotmail (1996) made webmail popular; Gmail launched in 2004.



anatomy of an email address



PUSH: SMTP all the way to the receiver's server

PULL: client fetches

MIME lets mail carry attachments, images and Unicode (Bangla) text

an email's journey: push with SMTP, pull with POP3 or IMAP

	SMTP	POP3	IMAP
Job	**send**, relay	download mail	read on the server
Port (secure)	25 (587, 465)	110 (995)	143 (993)
Copy on server	n/a	usually **deleted**	**kept**, synced
Best for	outgoing mail	one device, offline	**many devices**

Field	Meaning
To	main recipient(s)
Cc	carbon copy: visible to all
Bcc	**blind** carbon copy: hidden
Subject	one-line topic
Reply all	sender + every To/Cc
Forward	send a copy to someone new

Exam trap: **Bcc** recipients are invisible to everyone else, and "Reply all" does **not** reach them. SMTP is for sending only; reading mail needs POP3 or IMAP.

Memory trick: **S**MTP = **S**end. **P**OP = **P**ull & **P**urge. **I**MAP = **I**n place (mail stays on the server).

Worked example ✍️: An officer reads office mail on a desktop and a phone and wants both to show the same read/unread state. Answer: configure **IMAP** (port 993 with TLS) for receiving and SMTP (587) for sending. POP3 would download each message to one device and remove it from the server.

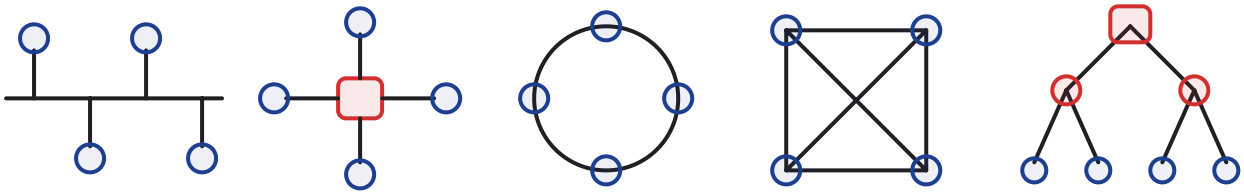
9 Computer Networks: Types

Network = two or more devices linked to **share data and resources**. Layers and protocols: Chapter 9.

Type	Full form	Span	Example
PAN	Personal Area Network	≈ 10 m	phone ↔ earbuds (Bluetooth), USB
LAN	Local Area Network	building	office Ethernet, Wi-Fi (campus: CAN)
MAN	Metropolitan Area Network	a city	city fibre / cable-TV net
WAN	Wide Area Network	country+	**Internet = largest WAN**

Internet = public · **Intranet** = private, staff only, built with web technology · **Extranet** = intranet opened to chosen partners. **Client–server**: a central server (bank, office) · **peer-to-peer**: all equal, no server (BitTorrent).

10 Network Topologies



Bus
one backbone

Star
hub / switch

Ring
token passes

Mesh
all-to-all

Tree
stars in levels

Topology	Plus	Minus
Bus	least cable, cheap, easy	backbone break = all down
Star	**most used in LANs**	hub/switch = single point of failure
Ring	no collisions (token)	one break stops it (except dual ring)
Mesh	**most reliable**, redundant paths	most cable and ports, costliest
Tree	scalable, hierarchical	root failure cuts off branches

Cable counts (n devices):
Full mesh: **n(n − 1)/2** links, n − 1 ports each
Star: n links · Ring: n links
Bus: 1 backbone + n drop lines

Worked example ✍️: Full mesh of 6 PCs: 6 × 5 / 2 = **15** cables, **5** ports per PC. For 10 PCs: 10 × 9 / 2 = **45**.

Memory trick: **B**us = **B**ackbone, **S**tar = **S**witch in the centre, **R**ing = **R**elay the token, **M**esh = **M**aximum links.

Exam trap: A star wired to a **hub** behaves logically like a bus (every frame goes to all). "Most reliable" = mesh; "most common" = star.

11 Transmission Media & Modes

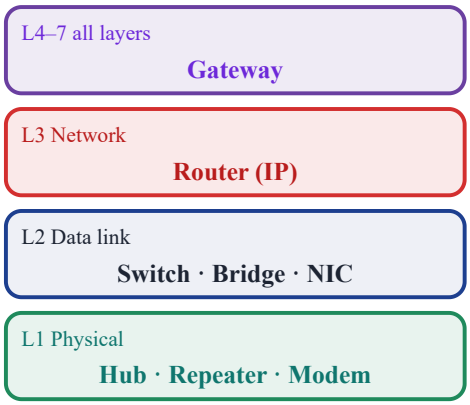
Medium	Key facts
Twisted pair	UTP/STP; Ethernet, **RJ-45** (phone RJ-11); ≤ 100 m per run
Coaxial cable	cable TV, old Ethernet; BNC connector; better shielding
Optical fibre	light by **total internal reflection**; highest bandwidth, no EMI
Wireless	radio (Wi-Fi) · microwave (line of sight) · infrared
Satellite	geostationary at **35,786 km**; VSAT; long delay

Mode	Example
Simplex: one way	TV broadcast
Half-duplex: turns	walkie-talkie
Full-duplex: both	telephone

> **Worked example ✍️:** 100 MB file at 20 Mbps: 100 × 8 = 800 Mb; 800 / 20 = **40 s**. Speeds are in **bits**/s; file sizes in **bytes**.

12 Network Devices

Device	What it does
NIC · repeater	has the MAC · boosts signal
Hub	copies data to **all** ports
Bridge	joins 2 LAN segments by MAC
Switch	sends to the right port (MAC)
Router	joins networks, routes by **IP**
Gateway	joins **different protocols**
Modem	digital ↔ analog signal
Access point	joins Wi-Fi clients to a LAN



which layer each device works at

13 Ways to Connect

IEEE: 802.3 Ethernet · **802.11 Wi-Fi** · 802.15.1 Bluetooth · 802.16 WiMAX. **Wired:** dial-up (≤ 56 kbps) → **ADSL** (download > upload) → cable → **FTTH**. **Mobile:** 2G GSM (voice, SMS) → 3G (broadband) → 4G **LTE** (all-IP) → 5G (low latency, IoT).

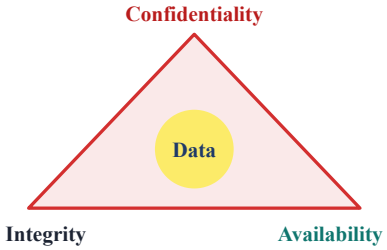
> **IT link:** A typical office or home line: fibre **ONU** (or modem) → **router** (NAT, DHCP server, firewall) → **switch** → PCs, with an **access point** for Wi-Fi. A home "Wi-Fi router" is router + switch + access point in one box.

Memory trick: **H**ub **H**ollers to all; **S**witch **S**elects by MAC; **R**outer **R**outes by IP.

Exam trap: "Wi-Fi" has **no official** full form ("Wireless Fidelity" is a slogan). 1 Mbps = 10^6 bits/s, not 2^{20}.

14 Cyber Security Basics

Cyber security: protecting systems, networks and data from attack and unauthorised access. Goals = **CIA triad**: **Confidentiality** (only authorised readers: encryption) · **Integrity** (no unnoticed change: hashes, digital signatures) · **Availability** (service up when needed: backups; DDoS attacks it).



Malware	How it behaves
Virus	attaches to a **host** file; spreads when that file runs
Worm	**self-replicates** over the network, no host or user action
Trojan horse	looks useful, hides a harmful payload; does not self-replicate
Ransomware	encrypts files, demands payment (WannaCry, 2017)
Spyware · keylogger · bot	spies · records keystrokes · zombie PC in a botnet

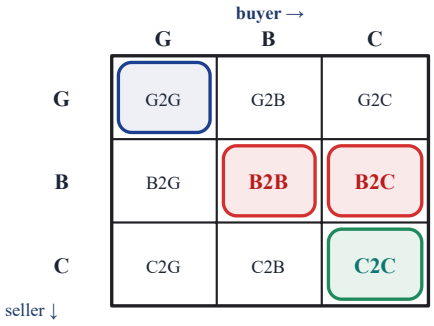
Attacks: phishing (fake mail/site; vishing = voice, smishing = SMS) · DoS/DDoS (flooding) · man-in-the-middle · SQL injection · social engineering (fake "bank officer" asks for the OTP) · brute force. **Defences:** firewall, antivirus + patches, strong passwords + **2FA/MFA**, HTTPS/VPN encryption, backups, awareness. **History:** Creeper 1971 · **Brain 1986**, first PC virus (Pakistan) · Morris worm 1988.

Bangladesh: Feb 2016 Bangladesh Bank heist: fraudulent **SWIFT** orders, about **US$81 million** lost. Laws: ICT Act 2006 → Digital Security Act 2018 → Cyber Security Act 2023 → Cyber Security Ordinance 2025. National CIRT: BGD e-GOV CIRT (BCC).

Memory trick: A **virus** rides a host, a **worm** crawls alone, a **Trojan** wears a disguise.

Exam trap: Firewall ≠ antivirus: a firewall filters **traffic**, antivirus scans **files**.

15 E-commerce



E-commerce = buying and selling over electronic networks with **online payment** (**m-commerce** on mobiles). **B2B** wholesaler → retailer (largest by value) · **B2C** shop → customer (Daraz) · **C2C** person → person (Bikroy) · **C2B** freelancer → company · **G2C** tax return, passport · **G2B** e-GP tenders. **Payment:** card via a payment gateway, internet banking, **MFS** (bKash 2011, Nagad, Rocket), cash on delivery.

> **Memory trick:** Read it as **seller 2 buyer**: the first letter provides, the last receives.

Night-Before-the-Exam Sheet

Fact sheet:

ARPANET 1969 · email 1971 (@)

TCP/IP: Cerf & Kahn · switch 1 Jan 1983

WWW: Berners-Lee, CERN, 1989

Mosaic 1993 · Google 1998

IPv4 32 bit · IPv6 128 bit · MAC 48 bit

Loopback 127.0.0.1 · private 192.168.x.x

DNS: name → IP (53) · Telnet 23

FTP 21 · SSH 22 · SMTP 25 · IMAP 143

HTTP 80 · HTTPS 443 · POP3 110

Mesh n(n−1)/2 links · star most common

Hub L1 · switch L2 · router L3

802.3 Ethernet · 802.11 Wi-Fi · 802.15.1 BT

BD: VSAT 1996 · SEA-ME-WE 4 2006

CIA = confidentiality, integrity, availability

Full forms that are asked

WWW World Wide Web

URL Uniform Resource Locator

ISP Internet Service Provider

DNS Domain Name System

HTTP HyperText Transfer Protocol

HTML HyperText Markup Language

TCP Transmission Control Protocol

SMTP Simple Mail Transfer Protocol

POP Post Office Protocol

IMAP Internet Message Access Protocol

FTP File Transfer Protocol

DHCP Dynamic Host Configuration Protocol

ARPANET Advanced Research Projects Agency Network

ICANN Internet Corp. for Assigned Names & Numbers

MODEM Modulator–Demodulator

VoIP Voice over Internet Protocol

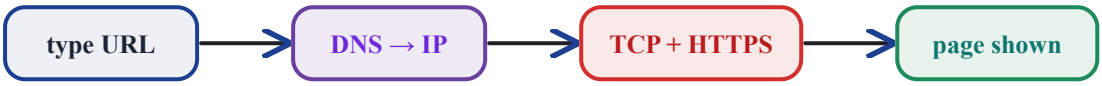
MIME Multipurpose Internet Mail Extensions

IIG International Internet Gateway

Top traps: Internet ≠ WWW · email (1971) is older than the Web (1989) · DNS ≠ DHCP · Bcc is hidden, Cc is not · POP3 deletes, IMAP keeps · hub ≠ switch · Mbps (bits) ≠ MBps (bytes) · Wi-Fi has no official full form · firewall ≠ antivirus · Trojan does not self-replicate

Bangladesh in one line: Internet 1996 (VSAT) · SEA-ME-WE 4 2006 (Cox's Bazar) · SEA-ME-WE 5 2017 (Kuakata) · 3G 2012 · 4G 2018 · BS-1 2018 · ICT Act 2006 · UDC 2010 · 999 (2017) · Bangladesh Bank heist 2016

All tricks together: DIKW pyramid · 69-83-89 spine · DNS = phone book, DHCP = hotel desk · the 20s count up (21, 22, 23, 25) · Send / Pull & Purge / In place · Backbone, Switch, Relay, Maximum · Hub hollers, Switch selects, Router routes · virus rides, worm crawls, Trojan disguises · seller 2 buyer



what happens when you open a website, in four boxes

✓ Practice MCQs (exam pattern, 10 marks)

1. Who invented the World Wide Web?
(a) Vinton Cerf (b) Ray Tomlinson
(c) Tim Berners-Lee (d) Bill Gates

2. How many bits are in an IPv6 address?
(a) 128 (b) 32
(c) 64 (d) 48

3. The main job of DNS is to:
(a) assign IPs (b) send email
(c) encrypt traffic (d) names → IPs

4. Default port of HTTPS:
(a) 80 (b) 443
(c) 21 (d) 25

5. Which protocol keeps mail on the server, synced across devices?
(a) SMTP (b) POP3
(c) IMAP (d) FTP

6. Links in a full mesh of 8 computers:
(a) 8 (b) 16
(c) 56 (d) 28

7. Which device routes packets between networks by IP address?
(a) Router (b) Hub
(c) Repeater (d) Switch

8. Malware that spreads over a network with no host file:
(a) Trojan (b) Worm
(c) Adware (d) Virus

9. A customer buying from Daraz is:
(a) B2B (b) C2C
(c) G2C (d) B2C

10. Bangladesh joined SEA-ME-WE 4 in:
(a) 1996 (b) 2006
(c) 2010 (d) 2017

Answer key → 1-c · 2-a · 3-d · 4-b · 5-c · 6-d · 7-a · 8-b · 9-d · 10-b

Written question: Distinguish between the Internet and the World Wide Web. Explain, step by step, what happens when a user types https://www.example.gov.bd in a browser. 8 marks

1. **Internet** = network of networks on TCP/IP (infrastructure); **WWW** = hypertext service on it (HTML, HTTP, URL). Email and FTP use the Internet but are not the Web. 2
2. Browser parses the URL: scheme https, host www.example.gov.bd, port 443. 1

3. **DNS**: the resolver asks root → .bd TLD → authoritative server and returns the IP. 2
4. **TCP** connection to IP:443, then a **TLS** handshake (certificate checked, keys agreed). 1
5. **HTTP GET**; server replies 200 OK with HTML; browser renders the page. 2

Before the exam ✓

- ✓ 1969 → 1983 → 1989 with names
- ✓ Parts of a URL and an email address
- ✓ Ten ports, IPv4/IPv6 sizes
- ✓ 5 topologies, mesh links
- ✓ BD: 1996, 2006, 2017, 2018

Also practise (written):

① LAN vs MAN vs WAN
② Hub vs switch vs router
③ Virus vs worm vs Trojan
④ E-governance in Bangladesh